PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Assignment 6: Stress-Testing The Parrot

Assignment

We'll be focused on large language models (LLMs) this week. Specifically, this assignment asks you to systematically probe an LLM of your choice to generate evidence that bears directly on the Bender et al. "stochastic parrot" argument, then write a structured analysis of what you find. The assignment is as follows:

1. Choose one of the general-use large language models that is available on Penn State's AI Studio; this might be GPT/ChatGPT, Google Gemini, or Claude.
2. Conduct a small, structured "interview" of an LLM, consisting of at about 15 prompts, organized into 2-3 categories chosen from the following menu of types:
    - *Grounding tests*: Questions that require real-world perception or embodied experience to answer correctly (e.g., sensory questions, questions about the student's own immediate context, questions about what is happening right now)
    - *Reference tests*: Questions where a correct answer requires knowing that words point to things in the world, not just that certain words follow other words (e.g., spatial reasoning, causal chains, simple physical experiments)
    - *Consistency tests*: Asking the same question multiple ways, or asking the model to contradict itself, to probe whether its "beliefs" are stable in any meaningful sense
    - *Anthropomorphism traps*: Prompts designed to elicit language about feelings, preferences, or self-awareness, followed by probing follow-ups.
3. Save the full transcript of your session, and submit it as an appendix.
4. Write a short paper, addressing the following things:
    (a) Description: What did you ask, and what did the model say? Summarize the most revealing exchanges.
    (b) Analysis: What do your results suggest about Bender et al.'s argument? Did the model behave like a stochastic parrot, or did some responses complicate that framing? Use Shanahan's points about anthropomorphic language to reflect on moments where you found yourself attributing understanding to the model.
    (c) Reflection: Did anything surprise you? Were there cases where fluency was so convincing that the parrot framing felt inadequate, and if so, what does that tell us?

Details

Your assignment should include a separate title page with your name, the name of the course, the name of the assignment, and the date on which it was completed / submitted. The body of the written assignment should be *double-spaced*, in a conventionally-sized (11 or 12 point) font. Each written part of the assignment

should be as short or long as you deem necessary; a rough target might be a total of 3-4 pages, *not* including the transcript. You need not cite any sources for this assignment, or include references, unless you choose to.

This assignment must be completed and submitted by Monday, October 26, 2026 at 11:59 p.m. ET. Submit your assignment by *creating / converting your assignment into PDF format*, and emailing it to Dr. Zorn (at `zorn@psu.edu`). Assignments received after the due date will not be accepted. This assignment is worth a possible 100 points.